

Mathematics A

Unit 1 • Chapter OL-T10 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 14 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T10 • Expertise Q16+ • 5 items • 14 marks

Chapter	Items	Marks	Starts
01. OL-T10 Surds	5	14	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Surds

Unit 1 • Expertise • 14 marks • 5 item(s)

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q23

4 marks

23 The diagram shows a cuboid with a square cross section.

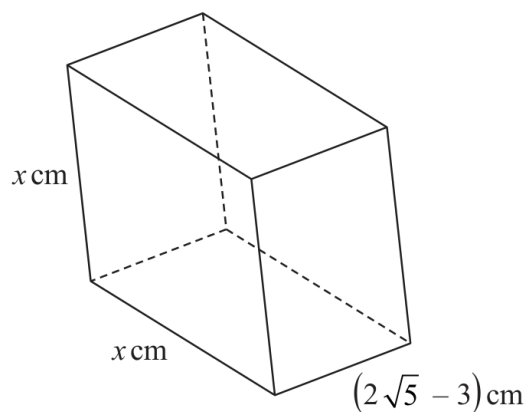


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Continue your method

4 marks

WORKING SPACE

4MA1-2H Q23

4 marks

- 23 The diagram shows a cuboid with a square cross section.

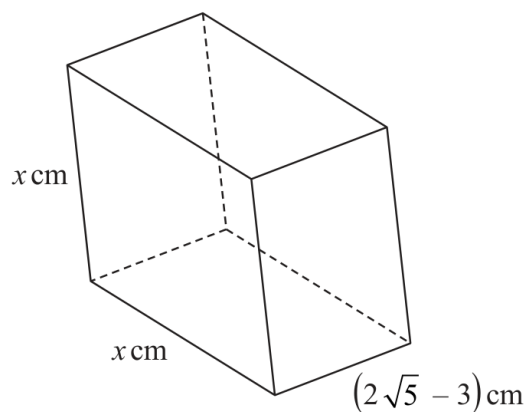


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The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

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Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x =$

Surd volume and a square cross-section**4 marks****Form x squared**

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5} \Rightarrow x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}.$$

Rationalise

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{11} = 9 + 4\sqrt{5}.$$

Recognise the square

$$9 + 4\sqrt{5} = (2 + \sqrt{5})^2.$$

Take the positive length

$$x = 2 + \sqrt{5}.$$

Final answer

$$2 + \sqrt{5} \text{ cm}$$

4MA1-1H Q17 M01

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

WORKING SPACE

Express a surd in the required form

1 marks

M01 – Express a surd in the required form**Factor inside the root**

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}.$$

Final answer

$$n = 5$$

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

WORKING SPACE

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

Chapter: Surds

3 marks

Q16 – Chapter: Surds

Family: Simplify and collect surds • Variant: Add or subtract like surds

Method

$$12/(\sqrt{2} - 1) = 12(\sqrt{2} + 1)/[(\sqrt{2} - 1)(\sqrt{2} + 1)] = 12\sqrt{2} + 12.$$

Why: Multiply by the conjugate to rationalise the denominator.

Method

$$(\sqrt{2})^5 = 4\sqrt{2}, \text{ so the left side is } 12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12.$$

Why: Use powers of $\sqrt{2}$ and collect like surds.

Method

Since $\sqrt{32} = 4\sqrt{2}$, the right side $2\sqrt{32} + 12$ is also $8\sqrt{2} + 12$; therefore the identity is shown.

Why: Simplify both sides to the same exact surd form.

Final answer

shown identity: Both sides simplify to $8\sqrt{2} + 12$

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

Worked solution

January 2020 | Question 17 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1. $\sqrt{12} = 2\sqrt{3}$, so $6 + 2\sqrt{12} = 6 + 4\sqrt{3}$.

2. Square: $(6 + 4\sqrt{3})^2 = 84 + 48\sqrt{3}$.

3. Factor 12: $84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$.

Final answer: $12(7 + 4\sqrt{3})$

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

19	$\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}}$ or $\frac{6}{3-\sqrt{7}} \times \frac{-3-\sqrt{7}}{-3-\sqrt{7}}$			M1
	$\frac{6(3+\sqrt{7})}{3^2-7}$ or $\frac{6(3+\sqrt{7})}{2}$ or $\frac{6(-3-\sqrt{7})}{-3^2+7}$ or $\frac{6(-3-\sqrt{7})}{-2}$			M1 (numerator may be expanded or denominator may be 4 terms which need to be all correct)
		$9+3\sqrt{7}$	3	A1 dep on M2 for $9+3\sqrt{7}$ or $3(3+\sqrt{7})$ from correct working
				Total 3 marks

Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

Method

$$(6)/(3-\sqrt{7})$$

Rationalise the denominator:

$$(6)/(3-\sqrt{7}) \times (3+\sqrt{7})/(3+\sqrt{7})$$

$$=(6(3+\sqrt{7}))/((9-7))$$

$$=(6(3+\sqrt{7}))/2=3(3+\sqrt{7})$$

$$=9+3\sqrt{7}$$

Answer: $9+3\sqrt{7}$